Assignment 2

A) Setup

1. Create a file in your terminal using nano named lab_2.txt
2. Copy the following content to the file

2024-09-23 12:00:00 INFO Process started
2024-09-23 12:00:15 DEBUG Checking for existing log files to prevent overwriting
2024-09-23 12:00:45 INFO Log file initialized: /var/log/lab_process.log
2024-09-23 12:01:00 DEBUG Initializing variables and configurations
2024-09-23 12:01:30 INFO Setting default values for parameters
2024-09-23 12:02:00 INFO Loading user preferences from /home/user/.config/preferences.yaml
2024-09-23 12:02:30 INFO Reading configuration files from /etc/config.yaml
2024-09-23 12:03:00 INFO Configuration loaded successfully
2024-09-23 12:03:30 DEBUG Verifying configuration settings for correctness
2024-09-23 12:04:00 INFO Connecting to the database at db.example.com
2024-09-23 12:04:30 INFO Database connection established successfully
2024-09-23 12:05:00 DEBUG User input received
2024-09-23 12:05:45 INFO User requested data analysis on input files
2024-09-23 12:06:00 DEBUG Validating user permissions for data access
2024-09-23 12:07:00 INFO Checking if user has read access to /data/input/
2024-09-23 12:08:00 WARNING User permissions insufficient for accessing /data/private/
2024-09-23 12:08:30 INFO Suggesting user request access to restricted directories
2024-09-23 12:09:00 INFO Executing fallback data retrieval method
2024-09-23 12:10:00 WARNING Memory usage is high
2024-09-23 12:10:30 INFO Monitoring memory usage patterns
2024-09-23 12:11:30 INFO Attempting to optimize memory allocation
2024-09-23 12:12:00 ERROR Could not find file /path/to/file
2024-09-23 12:12:30 INFO Logging error to error log: /var/log/lab_process_error.log
2024-09-23 12:13:00 INFO Searching for alternative file paths
2024-09-23 12:13:30 INFO Checking /data/backup/ for necessary files
2024-09-23 12:14:00 INFO File not found in alternative paths; user notified
2024-09-23 12:14:30 INFO Preparing to execute search commands
2024-09-23 12:15:00 INFO Executing command: grep -i "test" file.txt
2024-09-23 12:15:30 DEBUG Preparing parameters for grep command execution
2024-09-23 12:16:00 INFO Command executed with parameters: grep -i "test" /data/input/file.txt
2024-09-23 12:16:30 DEBUG Processing output of grep command

2024-09-23 12:17:00 INFO Capturing output lines that match the search criteria
2024-09-23 12:18:00 INFO Results of grep command:
2024-09-23 12:18:30 INFO Matching line: "This is a test line in the file."
2024-09-23 12:19:00 INFO Additional matching line: "Another test case found."
2024-09-23 12:20:00 INFO Command completed successfully
2024-09-23 12:21:00 INFO Aggregating results for reporting
2024-09-23 12:21:30 INFO Preparing summary report for the user
2024-09-23 12:22:00 DEBUG Generating summary statistics
2024-09-23 12:23:00 INFO Total matches found: 2
2024-09-23 12:30:00 DEBUG Searching for patterns in files
2024-09-23 12:31:00 INFO Checking file integrity
2024-09-23 12:31:30 INFO Performing checksum verification on input files
2024-09-23 12:32:00 INFO File integrity check completed without issues
2024-09-23 12:32:30 INFO Logging integrity check results
2024-09-23 12:35:00 INFO Line count: 100
2024-09-23 12:36:00 INFO Calculating processing time
2024-09-23 12:36:30 DEBUG Processing time: 35 seconds
2024-09-23 12:37:00 ERROR An unexpected error occurred
2024-09-23 12:37:30 INFO Error details: Insufficient memory to complete operation
2024-09-23 12:38:00 INFO Attempting to free up memory
2024-09-23 12:38:30 INFO Running garbage collection to reclaim memory
2024-09-23 12:39:00 INFO Cleanup process initiated
2024-09-23 12:40:00 INFO Closing unused resources
2024-09-23 12:41:00 INFO Cleanup completed successfully
2024-09-23 12:42:00 INFO Retrying previous operation
2024-09-23 12:42:30 INFO Operation resumed successfully after cleanup
2024-09-23 12:43:00 INFO Finalizing report generation
2024-09-23 12:44:00 INFO Operation completed successfully after cleanup
2024-09-23 12:45:00 INFO End of log

3. **Important: Use the sed command to prepend all lines of the file with your Student ID**
   sed -i 's/^/STUDENT_ID /' lab_2.txt

B) grep

1. Extract all lines from lab_2.txt that contain the word "INFO." How many lines did you find?
2. Find and display all lines that include the term "ERROR" case insensitive.

3. Use grep to count how many times the word "user" appears in lab_2.txt. What are the matching lines?
4. Search for lines that contain the word "memory" or “connection” and print those lines along with their line numbers.
5. Extract all lines containing "file" and display the total number of matches.

C) sed

1. Replace all occurrences of the word "INFO" in lab_2.txt with "INFORMATION." Save the output to a new file called updated_log.txt.
2. Extract only the timestamp and message from lines that contain "WARNING" and write the output to a file named warnings.txt.
3. Remove all lines that start with "DEBUG" from lab_2.txt and save the cleaned output to cleaned_log.txt.
4. Change all occurrences of "user" to "student" in lab_2.txt. Save the modified file as modified_log.txt.
5. Use sed to insert the text "Log Entry:" at the beginning of each line in lab_2.txt and save it to labeled_log.txt.
6. Delete all lines that do not contain the word "successfully" and save the output to success_log.txt.

D) awk

1. Use awk to print the timestamps and messages from lab_2.txt, but only for lines where the severity level is "ERROR."
2. Count the total number of lines in lab_2.txt using awk and print the result.
3. Display only the messages from lines that contain "Executing command" and the corresponding timestamps.
4. Calculate and display the total number of "INFO" lines in lab_2.txt using awk.
5. Print lines from lab_2.txt where it has “memory”, along with the timestamps.
6. Filter and print lines that mention "access" and display their line numbers.

E) Tar

1. Create a parent directory with the structure:
    a. lab2/dir1/dir2/dir3
    b. lab2/dir1/dir4
2. Move all files to lab2/dir1/dir2/dir3
3. Archive and compress the folder using tar
4. Print contents of archived folder
5. Remove the original folder
6. Unarchive the tar.gz file

Files to Submit:

- A PDF summarizing your results.
- Screenshots of the command executions and history